

SUNBLAZERLITE



Empowering Off-Grid Communities



The SunBlazer LITE is a next-generation product that is an easy-to-install PV solar-based community charging station.

The next-generation product in the SunBlazer series, the IEEE Smart Village SunBlazer LITE, has scalable features designed to provide reliable sustainable energy solutions to off-grid communities around the world.

IEEE Smart Village aims to relate across boundaries, incubate new designs with community input, demonstrate sustainable local market enterprise, and educate through linked learning communities.

These goals draw principally on the work of IEEE engineers and partner educators to develop synergistic and holistic community solutions. SunBlazer LITE aims to service smaller villages cost-effectively at the same time introducing a simple scaling model with added reliability through redundancy. Numbers of systems can be deployed throughout a community to provide many kilowatts of power as the community grows its demands.

Simultaneously, a reciprocal learning training and education model will enrich the knowledge and experience of students, engineers, and development practitioners globally.



How it Works

SunBlazer LITE is a mobile solar 500W to 1,050W base station including 100-200Ah at 24V station batteries, portable battery packs (PBKs) for home lighting kits. Each PBK has an auxiliary outlet for other DC powered devices in the home such as cellphones or small entertainment devices.

Assuming an average three- to four-day return for recharge and two charging shifts for the 12Ah or 18Ah PBK batteries, with 6 chargers the basic unit with two four-hour charging shifts will support 36 home lighting systems of 2-4W LED bulbs. With 10 chargers (10-pack) it can support 60 12Ah units.

The solar panels are held in a stand-alone frame with adjustable solar angle. The heavy duty plastic box containing the electronics can rest on the feet of the frame, or on a concrete pad nearby. The assembly needs protection by a security fence, or alternatively the box could go indoors and the panels mounted in a roof.

All PBKs and stations will contain a monitoring and telemetry feature to record by battery serial number/lessee every time a PBK is charged.

For larger home or base loads either a SunBlazer II with expansion capacity or multiple SunBlazer LITEs with expansion capacity can be deployed.

IEEE Smart Village aligns with the IEEE mission to stand at the forefront of technological change for the betterment of humanity. We are seeking volunteers and financial contributions to expand this program so it can incorporate member and society interests in the cross-disciplinary technology communities of IEEE worldwide.

If you are interested in assisting with this project, please visit ieee-smart-village.org or contact Michael Deering, IEEE Development Office, at +1 732-562-3915 | m.deering@ieee.org

SunBlazer LITE UNIT SPECIFICATIONS



SPECIFICATIONS	MECHANICAL
Dimensions	Transport 70"L x 48"W x 48"H Setup 102"L x 96" W x 90"H (PV panel pivot point)
Weight	905 lbs. 410 kilos - 2 solar panels, 2 station batteries
Sun Angle Adj.	-45 to +75 deg.
Terrain Requirement	Flat
Frame Construction	1/8" thick stainless steel box tubing, #4 finish
Enclosure Construction	High-density polyethylene (HDPE) plastic, powder-coated steel hinges
Mobility	Self-contained shipping crate
Security	Electronics and station batteries are accessible only by operator
Typical Setup Time	1 hour with 2 people



SPECIFICATIONS	ELECTRICAL
Solar Panels	Base Unit: 2-265 watt panels for 530 watts total Expanded Unit: 3-300W panels for 900W total
Solar Generation (2 or 3 panels)	Base Unit: 2.9 kWh/day in equatorial regions Expanded Unit: 4.9 kWh/day
Station Batteries	Base Unit: 2-12V 100Ah FLA flooded lead acid, 24 volt bus Expanded Unit: 2-12V 200Ah FLA, 24V bus
Solar Charge Controller	3 stage PWM, temperature compensated, automatic load disconnect if batteries are <50% charged and re-connection upon charge
Sub Chargers for Universal Portable Battery Kits (UPBK's)	6 charging channels for 12V 12 or 18 Ah UPBK batteries, or 10 charging channels for 10-12Ah units simultaneously. Each channel has 3 stages with LED indication of completed charge. Each channel has on-demand "restore charge" as needed to help extend cycle life
UPBK Charge Time/Capacity	4-hour charge time with capacity to support 2 shifts of 6-10-12Ah or 6-18Ah UPBK's per solar day
AC Output (optional)	400 watts at 230 VAC 50 or 60 Hz (expandable upon request)
AC Input (optional)	230 VAC battery charger if standby generator or grid AC is accessible



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For more information on IEEE Smart Village, visit: ieee-smart-village.org